

Assistant	CookSim							VHSim							ScreenSim						
	Interaction quality							Interaction quality							Interaction quality						
	In-time success	Factual grounding	Situational relevance	Actionable guidance	User intent uptake	Guidance conciseness	Overall	In-time success	Factual grounding	Situational relevance	Actionable guidance	User intent uptake	Guidance conciseness	Overall	In-time success	Factual grounding	Situational relevance	Actionable guidance	User intent uptake	Guidance conciseness	Overall
Frontier API models																					
Gemini 3.8 Flash	30.6	<u>82.7</u>	79.6	68.1	76.2	89.0	<u>53.4</u>	21.4	<u>82.7</u>	83.0	67.6	62.9	79.0	46.2	78.5	92.0	96.5	90.5	100.0	98.2	77.8
GPT-6 Astra	<u>26.5</u>	86.5	89.4	78.5	97.1	<u>86.1</u>	63.5	15.8	76.9	84.0	56.6	79.0	72.8	34.5	90.4	88.4	<u>96.1</u>	<u>93.7</u>	60.0	95.2	75.5
Gemini 3.1 Pro	20.8	61.5	66.0	56.6	80.4	65.4	27.9	17.6	71.4	78.0	61.3	67.8	68.6	30.8	<u>86.3</u>	91.5	93.8	93.8	33.3	96.2	79.6
Real-time streaming API models																					
GPT-Realtime 2.1	21.2	80.9	<u>87.7</u>	<u>72.6</u>	76.2	58.1	49.5	19.0	82.1	87.6	<u>66.6</u>	<u>77.4</u>	54.1	37.1	66.7	92.1	95.6	90.3	<u>87.0</u>	84.5	76.8
Gemini 3.8 Live	11.9	74.7	78.2	58.0	77.7	86.0	39.9	18.1	89.9	76.4	62.3	52.3	80.9	<u>42.1</u>	71.9	94.0	91.3	92.2	81.0	<u>96.9</u>	82.6
Open-weight non-streaming VLMs																					
Muse Glimmer	23.8	59.2	75.3	56.8	65.0	41.3	27.3	15.7	70.3	<u>85.1</u>	62.6	75.9	45.2	26.4	44.1	80.4	87.5	81.8	65.0	75.2	54.8
Qwen3.8-27B	18.4	70.9	81.7	62.4	<u>85.9</u>	71.9	37.4	25.2	68.7	78.7	62.8	67.2	58.5	28.5	67.8	<u>93.1</u>	94.2	92.9	69.1	96.1	<u>82.0</u>
Kimi-VL-A3B-Thinking	15.2	34.2	39.4	29.1	39.0	26.0	5.7	<u>24.3</u>	56.4	60.0	42.4	37.1	30.8	12.0	9.6	71.5	58.5	54.7	37.5	58.9	30.1
InternVideo3-8B	14.5	65.7	45.3	35.0	41.7	68.9	20.9	18.2	64.0	61.7	44.9	24.4	44.4	18.4	0.4	70.8	49.3	42.6	27.5	68.5	23.2
Molmo2-8B	13.4	54.5	31.7	32.3	33.4	19.2	11.6	2.6	<u>72.2</u>	55.8	42.0	20.3	41.6	17.0	0.7	<u>81.5</u>	31.3	38.2	24.0	65.7	15.7
Open-weight streaming VLMs																					
Qwen3-Omni-30B	26.2	65.8	63.8	51.2	54.3	67.6	25.4	13.4	75.7	83.1	60.6	59.5	<u>79.1</u>	30.2	15.2	82.2	70.6	72.4	22.0	85.0	49.6
MiniCPM-o 4.5	13.7	55.4	44.7	33.5	46.5	51.8	14.6	15.4	61.7	58.5	35.5	39.1	29.4	16.2	2.6	75.0	54.8	50.0	49.1	66.4	31.3
Proact-VL	11.3	66.4	25.6	17.2	28.4	44.5	15.8	20.2	69.7	30.1	22.0	20.5	25.5	16.3	0.0	82.2	10.2	12.5	6.9	43.3	12.4
StreamingVLM	10.9	75.8	25.0	34.8	34.4	22.9	24.5	19.0	76.7	34.4	29.4	29.0	19.5	23.4	0.0	92.6	14.9	22.4	0.0	30.9	26.9

Table 3: **Visual assistance across three environments.** All scores are on a 0–100 scale ($\uparrow$). Each environment reports in-time success, five rubric dimensions, and overall interaction quality. Models share CookSim’s within-family success ordering. Bold and underline mark the best and second-best assistant in each metric column, respectively.

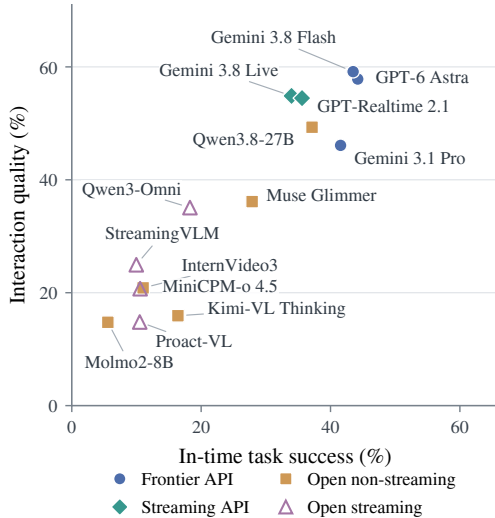


Figure 3: **Task success and interaction quality.** Each point represents one assistant, with equal weight given to the three environments. Colors and markers identify model families; model names are abbreviated. Higher values are better on both axes.

Table 3 reports in-time task success, five interaction-quality dimensions, and overall interaction quality in each environment. We average success rates over three independent runs and evaluate interaction quality using the full 16-item rubric on run 1. Rows follow CookSim’s success ranking within each model family, with the same model aligned across all three environments.

Figure 3 summarizes the two evaluation dimensions across environments, weighting each environment equally rather than pooling episodes. We retain per-environment scores because task difficulty and model rankings differ across domains.

Success measures engine-verified task completion within the deadline, whereas interaction quality evaluates the assistant’s responses throughout the rollout. The overall quality score is computed from all applicable items in the full rubric, including its hard requirements, rather than by averaging the five displayed dimensions. We report task success and interaction quality separately, without combining them into a single score.